

Enhancing Fetal Health Evaluation Through Lyrebird-Driven Ensemble Deep Transfer Learning on CTG Data

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Abstract—Monitoring the health of the fetus is an important component for the early detection and intervention of pregnant women. One standard way to assure the health of the fetus is through cardiotocography (CTG). CTG interpretation is typically subjective and inconsistent, which motivates the demand for such diagnostics model to be automated, accurate and interpretable. This article makes use of very sophisticated algorithms with the help of machine that assist automated diagnostic tools for categorizing CTG signals. Through learning, the healthcare practitioners can manage larger patient populations. With this motivation, we present a DL framework leveraging the CTG dataset to assess fetal health. Deep transfer learning (DTL) techniques extract important features that are then fused together to eliminate redundancy but keep informative features intact. Using the Lyrebird Optimization Algorithm (LO) model of feature selection is more efficient than the standard methods optimal balance between exploration and exploitation resulting which results into an optimal group of features while enhancing classification performance and reducing duplication. Many different classifiers exist in machine learning, including Boosting, Support Vector Machine, and Extreme Learning Machine (ELM). Furthermore, we develop an ensemble model (EffiDenseNet+LO) to enhance the accuracy. The experimental results show that the model achieves the best accuracy. The proposed framework is effective according to the results in which the accuracy has achieved up to 98.52% with ELM for the CTG fetal health monitoring mechanism. In comparison to other models, our model works more transparently and better. Moreover, it can provide practical decision-support for assessing fetal health in clinics.

Index Terms—Fetal Health Evaluation, DTL Techniques, Lyrebird Optimization Algorithm, Accuracy

I. INTRODUCTION

Fetal health monitoring is essential in prenatal care. It helps in early identification of abnormalities which may in-

terfere with the health of mother and baby. Cardiotocography (CTG) are continuous recordings of fetal heart rate (FHR) pattern and uterine contraction providing the most popular noninvasive measure of fetal status. Even so, the interpretation of CTG recording is still perceived to be highly interpretive and prone to inter-observer variability. The situation has resulted in an interest in the machine and deep learning solutions of automated classification of fetal-health [1].

In the last 10 years, researchers have used various methods to solve this problem. These methods include ensemble learning techniques, deep neural networks, classical machine-learning classifiers, and optimization-enhanced frameworks. Researchers used random forest, SVM MLP, and modified deep architectures in classifying CTG data as normal, suspicious and pathological. Use of everything from hybrid deep-learning-optimization approaches and Kolmogorov–Arnold Networks to the coupling of AlexNet with support vector machines has enhanced computational capability. Other studies have used IoT-enabled acquisition systems, stacking CNN ensembles for ultrasound-plane classification, and deep representations based on wavelets for CTG signal analysis [2]. Although these items may help the situation, the existing models mostly suffer from high dimensionality, class imbalance, insufficient feature representation, failure in model generalization across diverse CTG patterns and computational complexity. Due to the complexity of CTG signals and the variability of maternal–fetal conditions, no classifier or feature-learning method consistently outperforms the others. Which results, present-day solutions often find it difficult to keep the accuracy and robustness across multiple fetal-health classes, especially suspicious and pathological [3]. This modeling gap indicates

the need for an integrated, adaptive approach that will be capable of capturing the multi-dimensional structure of CTG data while optimizing for feature selection and model diversity.

This study proposes a unified framework called Enhancing fetal health evaluation through Lyrebird-Driven ensemble DTL on CTG Dataset which integrates the representational capability of deep transfer-learning architectures with the search efficiency of the Lyrebird Optimization Algorithm motivated by these challenges. The hybrid design intends to increase accuracy and stability as well as generalization. It provides a more reliable tool for fetal-health assessment, enabling better and timely decisions by the clinician.

II. LITERATURE SURVEY

The authors in [4] presented a prediction framework based on ensemble learning which would classify the health of the fetus, normal or suspicious or pathological. The cardiotocography (CTG) data is derived from fetal movements and fetal heart rate accelerations recorded during nonstress tests. A classification model that assigns three categories of fetal health was created using deep learning in [5]. The objective was to increase diagnostic efficiency and reliability through the use of deep neural architectures; for this they offer a DNN-based solution that leverages measurements from CTGs. In this study [6], machine learning approach which CTG data is analysed per labour stage was presented using different classical classifiers. The research in [7] presented an improved DNN model which efficiently recognises pathological and suspicious CTG traces that were often under-reported and has low complexity. Their approach speeds up computation by integrating AlexNet and SVM at the fully connected layers. Besides, the authors of [8] proposed a new predictive scheme using Kolmogorov–Arnold Networks (KANs) to model more elaborate interactions of CTG statistical features. KANs serve as a good alternative to neural transfer functions. The classification performance was improved. The validation was carried out on a publicly available CTG database labelled with fetal health condition. In [9], the authors proposed an automated fetal health assessment system that classifies the outcome of fetal health using deep learning. CTG data was first normalized using z-score and then deep features were extracted using AlexNet. Afterwards, feature selection was carried out using classification guided optimization which consists of k-nearest neighbours merged with War Strategy Optimization and Whale Optimization. The paper in [10] presented IoT-enabled fetal health detection and classification pipeline using the Mother Optimization Algorithm with DL (AFHDC-MOADL). The framework utilizes IoT sensors to acquire fetal data, pre-processes the data using KNN-based imputation and standard scaling, and leverages the Mother Optimization Algorithm for optimal feature subset selection to mitigate high dimensionality. In [11], the complementary strategy based on employing the Light Gradient Boosting Machine (LGBM) and CTG measurements applies SMOTE in the preprocessing stage to overcome the class imbalance for high-accuracy classification. According to survey in [12], recent

advances in deep learning for fetal ultrasound image interpretation were considered. This paper provides a brief yet very comprehensive overview for new and well-versed researchers. The fetal head, face, heart and abdomen core anatomical areas were the Focusing areas in this study. A novel CTG-net is proposed by [13] to deal with CTG datasets. A CNN model with complex Morse wavelet transforms was designed and applied to generate time–frequency FHR representation for Stage-I labor recordings with 94.1% accuracy [14]. An automated classification method of essential maternal–fetal ultrasound planes was proposed in [15], in which features extracted through AlexNet and VGG-19 (accompanied by global average pooling) were combined and subsequently classified by a multilayer perceptron. In [16], the authors build on this work and propose a stacked ensemble of deep CNNs: AlexNet, VGG-19, DarkNet-19 with Softmax and random forest decision layers; final predictions by majority voting. Ultimately, Dwivedi et al. [17] used the SMOTE method on CTG data to classify fetal health categories and to solve class imbalance. The model’s decision logic and relevant features were made available through the incorporation of explainable AI mechanisms.

Current models for fetal-health classification are defective because they use single learners, fixed feature extractors or optimizing methods not well-suited for the given CTG data, class imbalance and inter-patient variability. As per the No Free Lunch (NFL) theorem which states that no single algorithm can beat all others for all problems, the algorithms do not generally show reliable generalization for the minority fetal-health classes. Consequently, the need arises for a more adaptive and resilient framework. The proposed Lyrebird-driven ensemble integrates diverse deep transfer-learning models with the exploration–exploitation strengths of the Lyrebird Optimization Algorithm to address these gaps. It enhances feature representation and offers an enhanced and accurate solution for CTG based fetal health evaluation.

III. MATERIALS AND METHODS

A. *EfficientNet*

EfficientNet is a family of CNNs designed to maximise accuracy with low computational cost. The main strength derives from a compound scaling approach that simultaneously increases the depth, width and input resolution of the network, instead of tuning them independently. The architecture starts with a standard 3×3 convolution layer for primary feature extraction, followed by a sequence of MBConv blocks optimized through depth-wise separable convolutions with linear bottlenecks. Different MBConv variants utilize varying kernel sizes and expansion ratios, enabling effective hierarchical feature extraction as the network progresses through increasing channel dimensions and decreasing spatial resolutions. Each stage incorporates squeeze-and-excitation (SE) modules to enhance channel-wise attention and improve discriminative capability. The model concludes with a 1×1 convolution, global average pooling, and a fully connected layer to generate

compact and expressive feature representations. Its computational efficiency also makes it suitable for real-time or resource-constrained applications, including biomedical signal and ECG classification where maintaining low latency without compromising prediction quality is critical.

B. DenseNet

DenseNet-201 is a deep CNN composed of densely connected blocks, where every layer receives feature maps from all preceding layers, enabling stronger feature propagation, alleviating vanishing gradients, and reducing parameter redundancy. The network starts with a 7×7 convolution followed by a stride-2 max-pooling layer, and then proceeds through four dense blocks containing 6, 12, 48, and 32 layers respectively, each with a growth rate of $k = 32$. Within each dense block, feature maps are concatenated rather than added, leading to richer representation learning and more efficient feature reuse. Transition layers placed between dense blocks employ 1×1 convolutions with 2×2 average pooling to compress feature dimensionality and control model complexity. The final stage includes a 1×1 convolution, global average pooling, and a fully connected classification layer. Owing to its effective gradient flow, compact parameterization, and dense feature reuse, DenseNet-201 has shown strong performance in biomedical domains such as CTG signal classification, where capturing subtle waveform variations with high fidelity and robust generalization is essential.

C. Lyrebird Algorithm (LO)

The Lyrebird Optimization (LO) is a bio-inspired metaheuristic technique that models the evasive survival strategies of lyrebirds, where each individual alternates between escape and hide behaviours to balance exploration and exploitation during its search process. Consider a population of N candidate solutions represented as $X_i = (x_{i1}, x_{i2}, \dots, x_{id})$ in a d -dimensional search space, with fitness function $f(X_i)$ assessing the quality of each solution. At every iteration, the likelihood that a lyrebird adopts an escape action is evaluated through the escape probability

$$P_i = \frac{f_{\max} - f(X_i)}{f_{\max} - f_{\min} + \epsilon}, \quad (1)$$

where $f_{\max}$ and $f_{\min}$ denote the worst and best fitness values in the population respectively, and ϵ prevents division by zero. When an escape behaviour is triggered, the lyrebird performs a sudden displacement towards the global best or a randomly selected individual, expressed as

$$X_i^{t+1} = X_i^t + r_1 (X_{\text{best}}^t - X_i^t) + r_2 (X_{\text{rand}}^t - X_i^t), \quad (2)$$

where $r_1, r_2 \sim U(0, 1)$, X_{best}^t is the best solution of the current iteration, and X_{rand}^t is a randomly chosen individual. If the lyrebird chooses to hide instead, the movement becomes more exploitative, relying on a controlled adjustment towards the best-known position:

$$X_i^{t+1} = X_i^t + \alpha (X_{\text{best}}^t - X_i^t), \quad (3)$$

where the decaying adaptation factor α modulates the intensification capability and is defined as

$$\alpha = 2e^{-t/T}, \quad (4)$$

with t and T representing the current and maximum number of iterations, respectively. To preserve feasibility, all updated positions are restricted within their corresponding search boundaries using

$$X_{ij}^{t+1} = \min(\max(X_{ij}^{t+1}, LB_j), UB_j), \quad (5)$$

where LB_j and UB_j denote the lower and upper bounds of the j th dimension. For binary feature selection—such as fetal health classification from CTG datasets the continuous LOA search agents are converted to binary representations by mapping their values through a nonlinear transfer function

$$S(x) = \frac{1}{1 + e^{-x}}, \quad (6)$$

followed by a threshold-based discretization using

$$x_{ij}^{t+1} = \begin{cases} 1, & \text{if } S(x_{ij}^{t+1}) > r, \quad r \sim U(0, 1), \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

thus yielding a binary selection mask in which 1 denotes a chosen feature and 0 indicates exclusion. Through these mechanisms, LOA evolves its population by harmonizing rapid exploratory movements with fine-grained exploitative adjustments, making it effective for optimization problems including biomedical feature selection and classification tasks.

D. Proposed model

The suggested framework uses a deep learning-driven pipeline that processes the real CTG dataset for classification and health assessment of fetal signals. The raw CTG signals are processed using a wavelet based approach by decomposing each recording. This stage helps to suppress noise and enhance clarity of the signal. In addition, the transformed samples still contain essential physiological patterns needed for subsequent feature extraction.

After preprocessing, the features are learnt using EfficientNet and DenseNet. EfficientNet model can produce high-accuracy representations by using compound scaling without increasing model complexity. On the other hand, DenseNet connectivity allows every layer to receive feature maps from all layers preceding it. They both work on the CTGs which undergo wavelet transformation. Their features are kept separately to maintain the complementary information supplied by each architecture.

The two sets of features are then fused together by assigning weights to them based on the performance of each model. Variance thresholding is then used to remove redundant or minor features leaving only the most discriminative features. The Lyrebird Optimization (LO) algorithm is used for refining feature selection which is a probabilistic metaheuristic algorithm inspired by the foraging behaviour of a lyrebird. The LO iteratively selects a feature subset using a fitness

function based on the Balanced Accuracy. The best performing feature subset is selected after several iterations which helps to converge on the best. With the refined features, the system constructs an ensemble of classifiers, including SVM, AdaBoost, and ELM. The final decision is produced through a weighted averaging strategy, yielding a robust ensemble model optimized for CTG-based fetal health classification. The proposed model is presented using figure 1.

One effective method of feature fusion is the Weighted Feature Averaging Fusion combined with Variance Thresholding, where features extracted from DenseNet and EfficientNet are jointly utilized for CTG signal classification. In this approach, the feature vectors obtained from EfficientNet (FE) and DenseNet (FD) are fused through a weighted averaging strategy as expressed in Eq. (8), in which w_1 and w_2 denote the weight factors corresponding to the respective extracted features. The resulting fused feature vector f represents the final combined representation obtained after the fusion stage. The fusion operation is defined as

$$f = w_1 \cdot FE + w_2 \cdot FD, \quad (8)$$

where the weights satisfy the convex combination constraint shown in Eq. (9),

$$w_1 + w_2 = 1. \quad (9)$$

This constraint originates from the principle of convex combinations frequently applied in machine learning, ensuring that contributions from different transfer learning models are blended in a balanced and interpretable manner. The weights in Eq. (8) may be assigned using adaptive strategies, fixed scaling rules, software-based automatic weight tuning, or data-driven feature ratio approaches. In this work, both EfficientNet and DenseNet are treated with equal importance; therefore, the weights are set to $w_1 = w_2 = 0.5$ in accordance with Eq. (9).

ALGORITHM 1: PROPOSED CTG FEATURE FUSION AND LOA-BASED OPTIMIZATION FRAMEWORK

Input: CTG dataset

Output: Final performance metrics of the ensemble classifier

- 1) Load the original CTG dataset.
- 2) Let $D \leftarrow \text{Load_CTG}()$.
- 3) Perform wavelet-based preprocessing on each CTG sample to enhance morphological patterns.
- 4) Initialize an empty container WT_D for the transformed signals.
- 5) For every record $D[i]$:
 - a) Compute the wavelet representation: $WT_sample \leftarrow \text{Wavelet Transform}(D[i])$.
 - b) Append WT_sample to WT_D .
- 6) Extract deep-level descriptors from both CNN models:
 - a) $F_E \leftarrow \text{Extract Features}(\text{EfficientNet}, WT_D)$,
 - b) $F_D \leftarrow \text{Extract Features}(\text{DenseNet}, WT_D)$.
- 7) Fuse the deep descriptors through weighted feature averaging followed by variance-based filtering:
 - a) $F_{avg} \leftarrow \text{Weighted Average}(F_E, F_D)$,

- b) $F_{fused} \leftarrow \text{Variance Thresholding}(F_{avg}, \eta)$.
- 8) Use the Lyrebird Optimization Algorithm (LOA) to identify the optimal feature subset.
- 9) Initialize a population of candidate solutions: $L \leftarrow \text{Initialize Lyrebirds}()$.
- 10) Assign $\text{Best_Candidate} \leftarrow \text{null}$.
- 11) Set maximum iterations ($\text{MaxIter} = 100$) and iteration counter $t = 0$.
- 12) While $t < \text{MaxIter}$:
 - a) For each lyrebird L_i :
 - i) Evaluate the fitness using CTG classification accuracy with SVM: $\text{Fitness}[L_i] \leftarrow \text{Balanced Accuracy}(\text{SVM}, F_{fused}, L_i)$.
 - ii) If $\text{Fitness}[L_i]$ exceeds the current best, update: $\text{Best_Fitness} \leftarrow \text{Fitness}[L_i]$, $\text{Best_Candidate} \leftarrow L_i$.
 - b) Update all lyrebird positions using LOA's escape-hide behaviour model equations.
 - c) Increase iteration count: $t \leftarrow t + 1$.
- 13) Determine $\text{Selected_Features} \leftarrow \text{Best_Candidate}$.
- 14) Split the optimized features into training and testing sets with ratio 80:20.
- 15) Train classifiers from the set $\{\text{SVM}, \text{AdaBoost}, \text{ELM}\}$ on the training set.
- 16) For each classifier:
 - a) Train on the selected training portion.
 - b) Generate predictions for the test samples.
 - c) Store the classification outputs.
- 17) Apply weighted averaging ensemble strategies.
- 18) Compute and report the final performance metrics of the ensemble model.

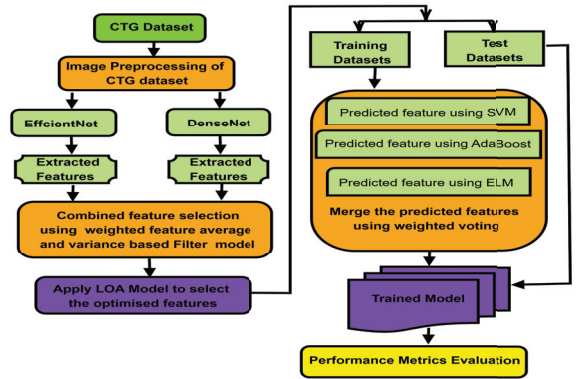


Fig. 1: Lyrebird-Driven Ensemble Deep Transfer Learning Model.

IV. EXPERIMENTAL DETAILS AND RESULT FINDING

A. Dataset Details

To evaluate the proposed model, cardiotocography dataset is used in the experimental analysis [18]. The used CTG

TABLE I: Performance Comparison of EfficientNet-based and Ensemble-based Models(%)

Model	ACC	MCR	PRE	REC	F1-S	SPE	FNR	FPR	MCC
EfficientNet + LO Models									
EfficientNet+LO+SVM	94.92	5.08	95.68	98.10	96.88	82.05	1.90	17.95	83.56
EfficientNet+LO+AbaBoost	96.46	3.54	96.75	98.68	97.70	89.36	1.32	10.64	90.10
EfficientNet+LO+ELM	97.45	2.55	97.97	98.64	98.31	93.88	1.36	6.12	93.16
DenseNet + LO Models									
DenseNet+LO+SVM	97.47	2.53	97.18	99.28	98.22	93.22	0.72	6.78	93.94
DenseNet+LO+AbaBoost	95.63	4.37	97.02	97.60	97.31	87.18	2.40	12.82	85.64
DenseNet+LO+ELM	97.98	2.02	97.67	100.00	98.82	86.67	0.00	13.33	92.01
EffiDenseNet + LO Models									
EffiDenseNet+LO+SVM	97.46	2.54	97.44	99.35	98.38	90.91	0.65	9.09	92.60
EffiDenseNet+LO+AbaBoost	98.48	1.52	98.75	99.37	99.06	94.87	0.63	5.13	95.18
EffiDenseNet+LO+ELM	98.52	1.48	98.05	100.00	99.02	94.23	0.00	5.77	96.12

dataset is publicly available at UCI Machine Learning Repository <https://archive.ics.uci.edu/ml/datasets/cardiocography>. The samples are taken from 2126 pregnant women. This dataset consists of 23 no of attributes to assess the FHR and UC of the women. Each labeled dataset having 30 minutes duration of recording taken by the obstetrics experts. CTG dataset is classified into three classes such as Physiological, Suspicious, and Pathological. The available is divided into training set (1643 records) and test set(483 records).

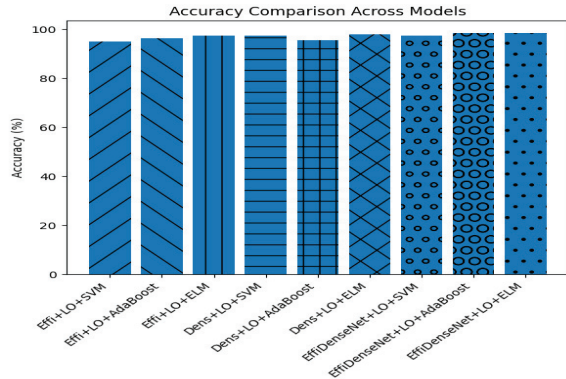


Fig. 2: Performance Analysis of the Proposed Model.

B. Results Analysis

The analysis of the nine configurations together shows some clear trends across the performance parameters, showing how the various models behave with respect to one another is presented in 2. The ensemble variants significantly outperform the EfficientNet models accuracy and misclassification rate. In fact, EffiDenseNet+LO+ELM achieves the highest accuracy at 98.52% and the lowest misclassification rate through EfficientNet. As a result, this shows that ensembling does impart measurable stability to the decision boundaries. In a similar manner, precision is similar with EfficientNet+LO+ELM has a large precision of 97.97%, while EffiDenseNet+LO+AbaBoost (98.75%), and EffiDenseNet+LO+ELM (98.05%). This result also indicates, the ensembles reduce false alarms better. Recall shows greater variation across configurations. Effi-

cientNet+LO+ELM and the two ELM-based ensemble models attain perfect recall (100%). Contrary to this, EfficientNet+LO+SVM receives the highest recall (99.28%) among its group but does not attain the highest of the ensembles. This shows that ensembles can recall all relevant instances even in difficult instances. The F1 scores follow the same trend with the variants of the ensemble again leading first is EffiDenseNet+LO+AbaBoost with a score of 99.06% and second is EffiDenseNet+LO+ELM with a score of 99.02%. Both scoring higher than all other Efficientnet-only models, which confirms an overall better balance between sensitivity and precision. The ensemble approach is advantageous due to its specificity and false-positive behavior. The EfficientNet model shows moderate to high specificity in most cases (82-94%). However, ensemble versions consistently push this bound upward. EffiDenseNet+LO+AbaBoost has the best specificity of 94.87% and lowest FPR of 5.13%. This illustrates the best discrimination against the negative class. Unlike the previous strategies, the first one highlighted the sensitivity strengths of the ELM-based models. Moreover, all ELM combinations (EfficientNet and EffiDenseNet) yield the lowest FNR values. Further, the model recorded several zeroes, showing its strength. When considered together, MCC provides the most clear-cut single measure of model reliability. EfficientNet+LO+ELM leads the pack with an MCC of 93.16%, but EffiDenseNet+LO+ELM enhances this significantly to 96.12%, the highest overall value of all the models, indicating that the ensemble architecture improves not just class-specific metrics but also the overall correlation of predictions to true. To sum up, a parameter-by-parameter comparison shows that while EfficientNet-based models already have solid baselines, the ensemble configurations constantly improve precision, reduce error rates and strengthen both sensitivity and specificity giving more trustable and balanced performances across all evaluation metrics. Below Figures 3 – 5 presents the ROC performance of all proposed models.

V. CONCLUSION

Using deep transfer learning with optimal feature-selection strategy along with CTG signal gives reliable results for

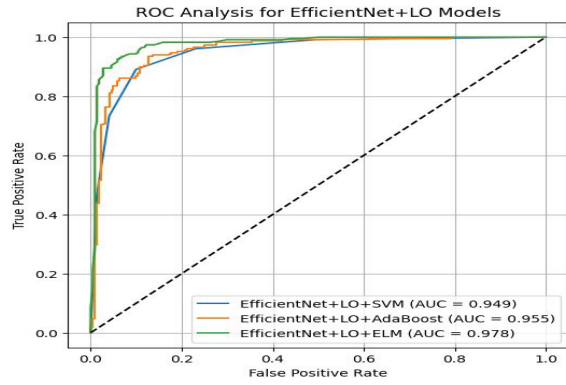


Fig. 3: Performance Analysis of the EfficientNet+LO Model.

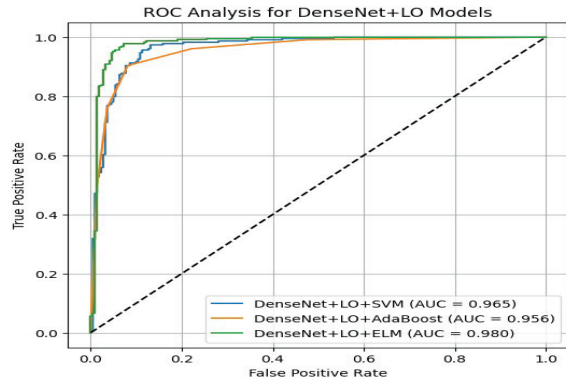


Fig. 4: Performance Analysis of the DenseNet+LO Model.

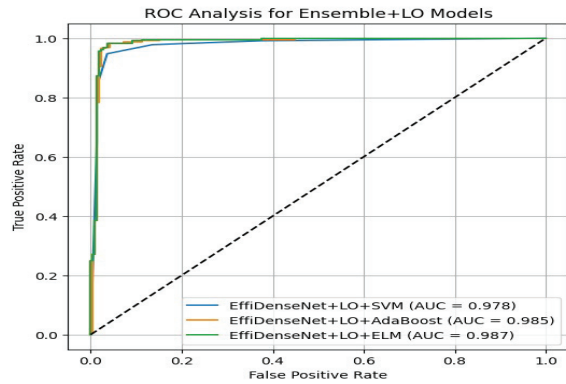


Fig. 5: Performance Analysis of the EffiDenseNet+LO Model.

assessment of fetal health shows this research work. The proposed framework reduces the redundant features and preserves informative features using the Lyrebird Optimization Algorithm from the high-level representations extracted. The performance of multiple classifiers, namely, SVM, AdaBoost, and ELM, confirmed that the chosen features improved stability and strength. The EffiDenseNet-based ELM model achieved the maximum accuracy of 98.52%. Using DTL-based feature extraction and LO-driven feature selection has the potential to create more robust and interpretable diagnostic tools. In overall, the proposed system is more consistent and transparent

than using the traditional manual CTG interpretation and has a strong opportunity for its use in every-day real-life situations to help the clinical decision-making support, particularly in fast-paced settings that require reliable and quick fetal health monitoring.

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